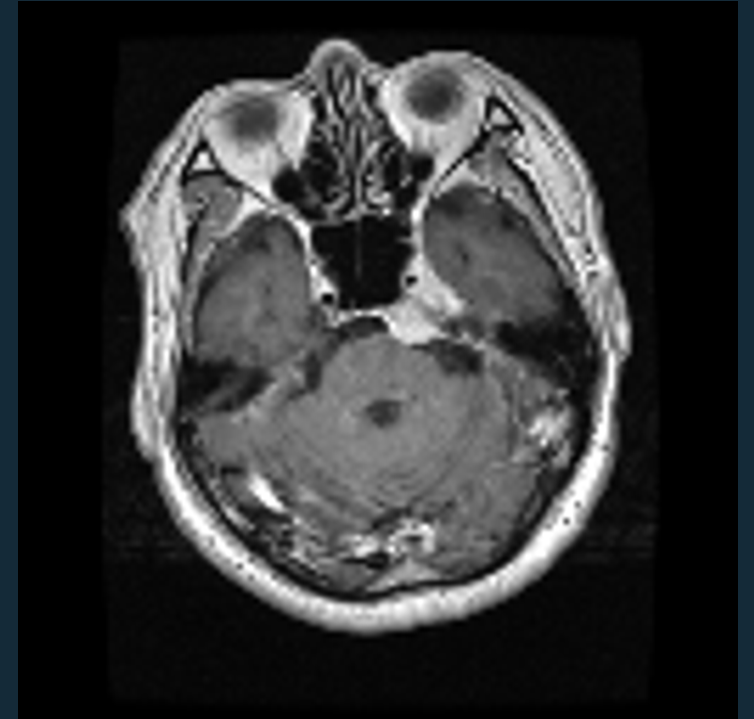


SoftROI-Boundary U-Net

A proposed architecture for
brain tumor segmentation

Project progress update
Small-subset experiment with two training seeds



Motivation from the base paper

The paper uses YOLOv5 to select a tumor region, then U-Net to segment the extracted patch.

Design choice	Base paper	Our proposal
Region selection	Detected bounding box	Learned soft ROI map
Image context	Cropped patch	Full image retained
Segmentation	Separate 2D U-Net	Shared network + boundary refinement

Hypothesis: soft guidance may reduce dependence on the quality of a hard crop.

Proposed architecture

Stage	Representation	Operation
Shared encoder	128 x 128 MRI to 16 x 16 features	Extract features at four scales
Soft region guidance	16 x 16 tumor probability R	Reweight skip features by $0.5 + R$
Decoder + boundary head	128 x 128 mask and boundary logits	Preserve context and predict edges
Uncertainty refinement	$u = 4p(1-p)$	Final logit = base logit + $u \times$ residual

492,167 parameters. The full proposal adds 0.83% over the shared baseline.

Ground-truth masks supervise training targets only. They never enter the forward pass.

Controlled experiment

Split	Patients	MRI slices
Training	48	398
Validation	12	117
Test	15	129

128 x 128 inputs. Three tumor classes. Two seeds per variant.
60 epochs, batch size 16, AdamW. No patient overlap.

Best checkpoint: validation patient Dice. Fixed threshold: 0.5. Test set used for final scoring.

Held-out segmentation results

Model	Patient Dice	Patient IoU	Boundary F1
Compact U-Net	59.79%	49.82%	60.71%
SoftROI	61.97%	52.04%	62.73%
SRB-U-Net	63.17%	53.76%	65.79%

+3.38 percentage points

SRB minus baseline in patient Dice. 95% paired interval: [-0.18, +6.74] points.

Preliminary result on 15 test patients. Paper benchmark values are not directly comparable.

Ablation and repeatability

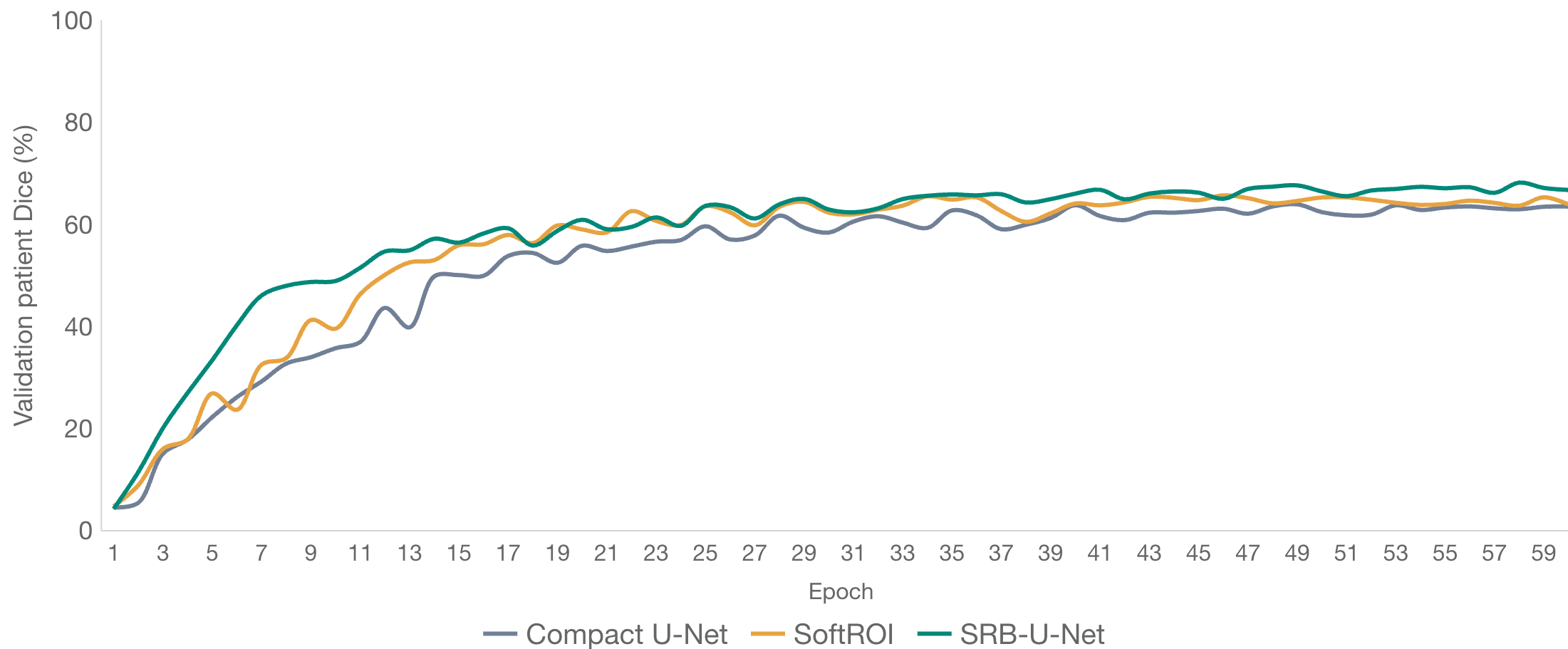
Model	Seed 17 Dice	Seed 29 Dice	Parameters
Compact U-Net	59.80%	59.78%	488,100
SoftROI	59.59%	64.36%	488,229
SRB-U-Net	63.29%	63.04%	492,167

Adding region guidance: +2.18 points in patient Dice.

Adding boundary/refinement: +1.19 points beyond SoftROI.

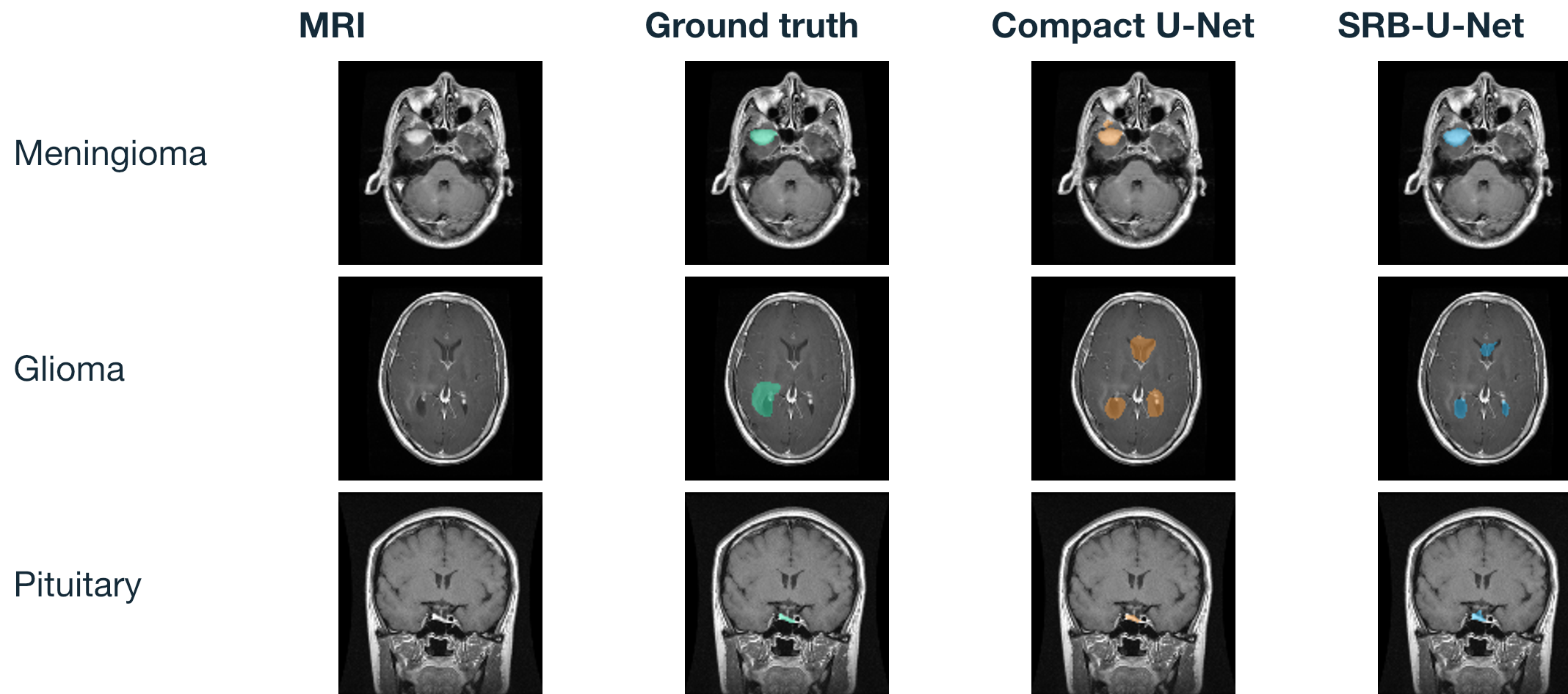
The current ablation tests modules plus their losses. A finer ablation is the next step.

Validation learning curves



Each curve averages two seeds. The training budget and checkpoint rule are the same for all variants.

Qualitative test examples



Fixed example rule: first held-out slice per class in manifest order. Predictions from seed 17.

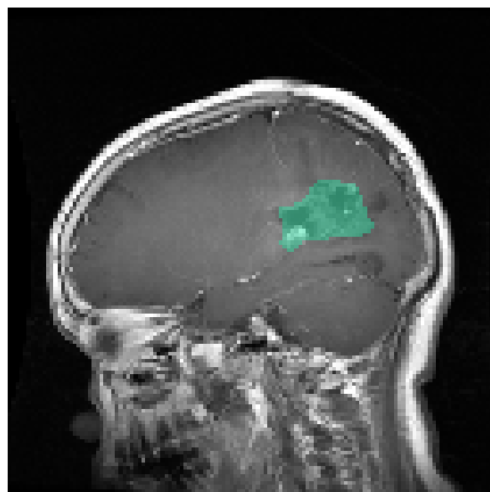
Observed failure case

Largest SRB regression relative to baseline: 2803.mat (seed 17)

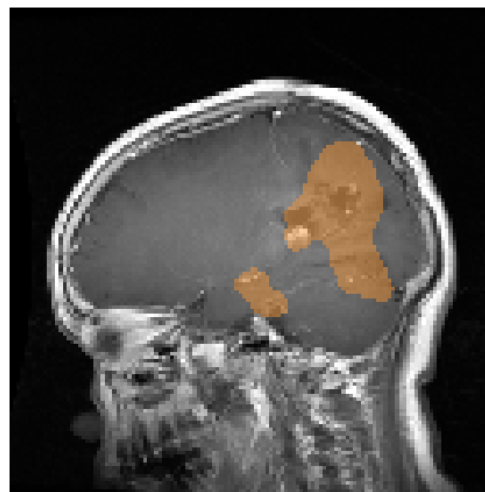
MRI



Ground truth



Baseline Dice 51.6%



SRB Dice 0.0%



The mean result does not guarantee a better mask for every slice.
Failure analysis will guide a fuller ablation and higher-resolution runs.

Current contribution and next experiments

Implemented: a soft ROI gate, boundary prediction, and uncertainty-weighted residual refinement in one compact network.

Next experiments

1. Repeat across patient folds and additional seeds.
2. Separate the effect of each module and auxiliary loss.
3. Increase resolution and reproduce the paper baseline.

Scope: a project architecture proposal and small-subset pilot. No claim of global novelty or clinical validation.

Appendix: additional measured outcomes

Outcome	Compact U-Net	SoftROI	SRB-U-Net
Meningioma patient Dice	68.78%	67.29%	66.84%
Glioma patient Dice	36.80%	37.82%	39.83%
Pituitary patient Dice	73.79%	80.80%	82.83%
Small-tumor slice Dice	62.99%	66.64%	68.66%
Tumor-type accuracy	86.82%	80.62%	87.21%

Two-seed averages. Small-tumor cutoff comes from training data. Classification is an auxiliary outcome.

Appendix: related work and evidence

Ahsan et al. (2024 online / 2025 issue)

Brain tumor detection and segmentation using deep learning

Oktay et al. (2018)

Attention U-Net: Learning Where to Look for the Pancreas

Xu et al. (2024)

Boundary guidance network for medical image segmentation

Li et al. (2022)

Category Guided Attention Network for Brain Tumor Segmentation in MRI

Source URLs are in speaker notes. Code, checkpoints, split manifest, and raw metrics accompany the deck.